**SCORING GUIDE FOR UTEC UCE PHY 2**
**PREPARED BY MR NYARUHUMA THOMAS 0703240480**

**GET YOURSELF MY UCE PHYSICS BOOK COVERING ALL THE NOTES, GIVING 20 SCHENARIO BASED ITEMS PER ELEMENT OF CONSTRUCT, HAVING 10 ANSWERED TESTS FOR YOU TO KNOW HOW TO PRESENT YOUR RESPONSES, HAVING FOURTEEN KEY PRACTICALS ANSWERED, UNEB PAST PAPERS**

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**535/2**
PHYSICS
**Paper 2**
Jul/Aug. 2026
**2 hours**

**UGANDA TEACHERS' EDUCATION CONSULT (UTEC)**

Uganda Certificate of Education

**Physics**

535/2

**(Practical)**

Time: 2 hours

**INSTRUCTIONS TO CANDIDATES:**

*This paper consists of* **two** *examination items.*

*Answer only* **one** *item in total.*

*Any additional items will not be scored.*

*Candidates are not allowed to start working with the apparatus for the first quarter of an hour. This time is provided to enable candidates to read the items thoroughly, check the required apparatus, and plan appropriately.*

*Graph paper will be provided.*

*Mathematical tables and non-programmable calculators may be used.*

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**Item 1**

The factory quality assurance office allows only SIZE D dry cells with lost voltage within the range of $0.052V - 0.468V$ , when a current of $0.52A$ is drawn from it to proceed to market. You are hired as a quality assurance manager and given a Size D cell from a particular consignment to determine whether it passes the test to proceed to market.

**Task:**

Carry out a scientific investigation to determine the fate of that dry cell.

**Hint:** Include the following in your report;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyze data (using graph) and recommend.

**Item 2**

An eye clinic officer requires convex lenses whose power lies between $9.52D - 10.53D$ for eye correction. The officer has a convex lens supplied to you but does not know whether it's of the required power.

**Task:**

Carry out a scientific investigation to help the officer.

**Hint**: Include the following in your report.

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyse data (using graph) and recommend.

**END**

# ITEM ONE

**Aim**

To determine the lost voltage by the cell in order for it to proceed to the market or not

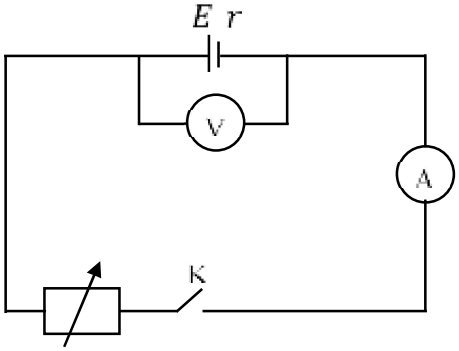
**Variables**

| | |
|---|---|
| Independent variable | Voltmeter reading, V |
| Dependent variable | Ammeter reading, I |
| Fixed variables | Temperature |

**Hypothesis**

The lost voltage lies between 0.052V-0.468V

**Procedures**




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**Steps**

a) The switch was closed
b) The rheostat was adjusted until the ammeter reading, I was 0.10A.
c) The voltmeter reading V was noted.
d) The steps a) to b) were repeated for I = 0.20, 0.30, 0.40, 0.50 and 0.60A.
e) A graph of V against I was plotted.
f) The slope s of the graph was obtained and the internal resistance r =-s
g) The lost voltage was calculated from $lost\ voltage = Ir\ or\ lost\ voltage = E - V$

**Possible Sources of errors and precautions**

| Source of error | precaution |
|---|---|
| Parallax errors when reading the ammeter and voltmeter | Placing the eye perpendicular the reading |
| Loose wire junctions | Making tight wire junctions |
| Change in temperature | Performing the experiment fast and in a cool environment |

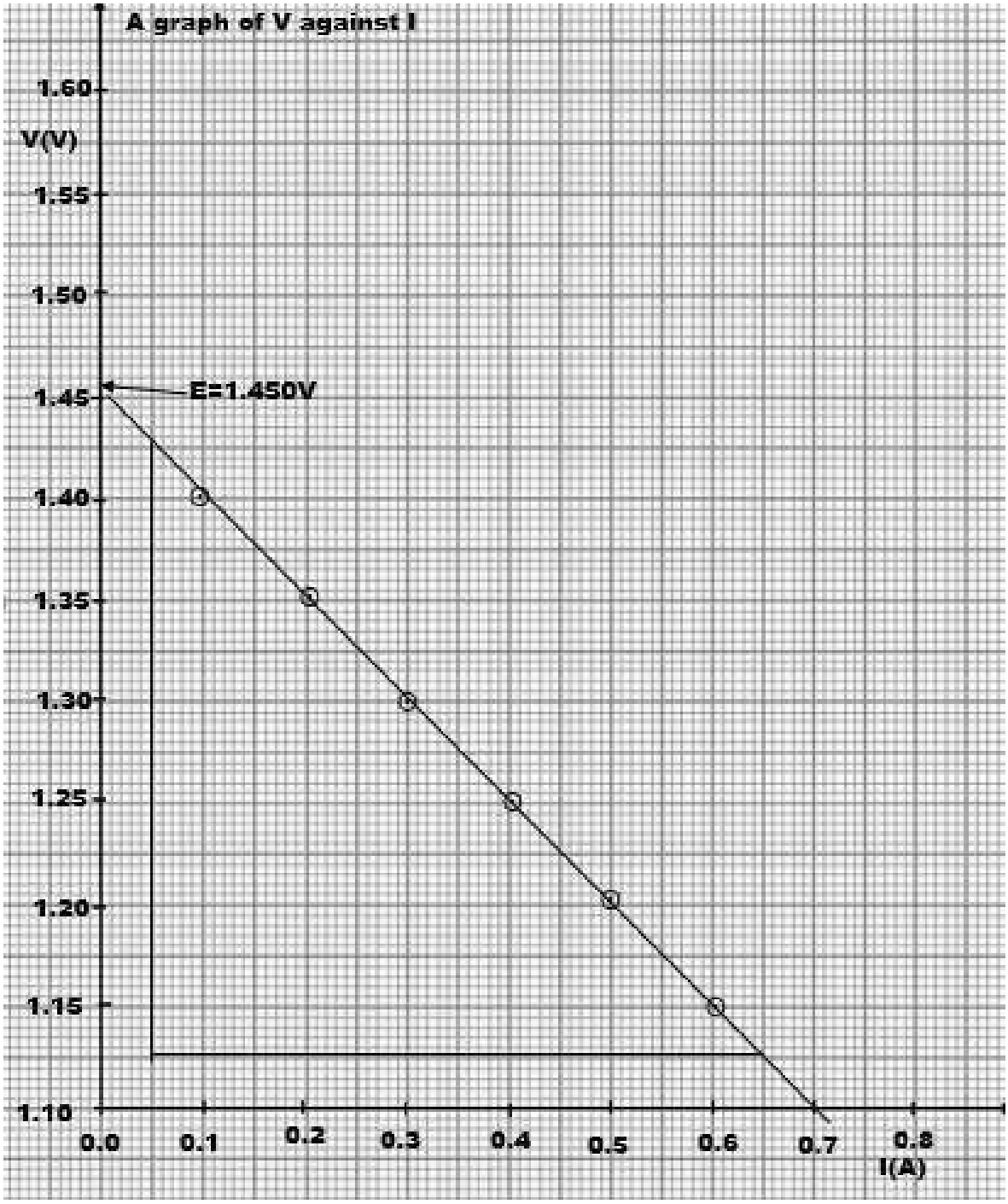
Table of results

| $I(A)$ | $V(V)$ |
|---|---|
| 0.10 | 1.40 |
| 0.20 | 1.35 |
| 0.30 | 1.30 |
| 0.40 | 1.25 |
| 0.50 | 1.20 |
| 0.60 | 1.15 |

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Slope S=$\frac{1.425-1.125}{0.05-0.65} = \frac{0.300}{-0.60} = -0.50\boldsymbol{\Omega}$ $\quad \boldsymbol{r = -s = 0.50\Omega}$

$lost\ voltage = Ir = 0.52x0.50 = 0.26V$

$or\ lost\ voltage = E - V = 1.450 - 1.19 = 0.26V$

**Conclusion**

The lost voltage is 0.26V hence the hypothesis is correct

**Recommendation**

The cell should proceed to the market

# ITEM TWO

**Aim**

To determine the power of the lens provided so as to determine whether it lies between 9.52D -10.53D in order to be used for eye correction

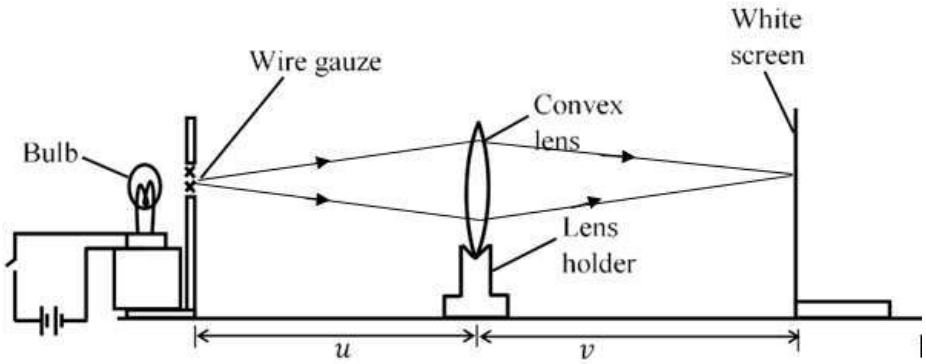
**Variables**

| | |
|---|---|
| Independent variable | Distance, $u$ from the wire gauze to the lens |
| Dependent variable | Distance, $v$ from the lens to the screen |
| Fixed variable | Light intensity, emf of the cells, material of the lens |

**Hypothesis**

The power of the lens lies between 9.52D -10.53D

**Procedures**




a) The distance, $u$ was adjusted to $u = 20.0\ cm$
b) **The switch was closed.**
c) The position of the white screen was adjusted until a clear image of the wire gauze is formed on it. d) The distance, $v$ was measured and noted.
e) The steps a) to d) were repeated for $u = 30.0, 40.0, 50.0, 60.0$ and $70.0\ cm$
f) The results are tabulated including values of $uv$ and $u + v$.
g) A graph of $uv$ against $u + v$ was plotted.
h) The slope of the graph was obtained.
i) The focal length f was determined from f=s
j) The power is calculated from $p = \frac{1}{f(m)}$

Sources of errors and precautions

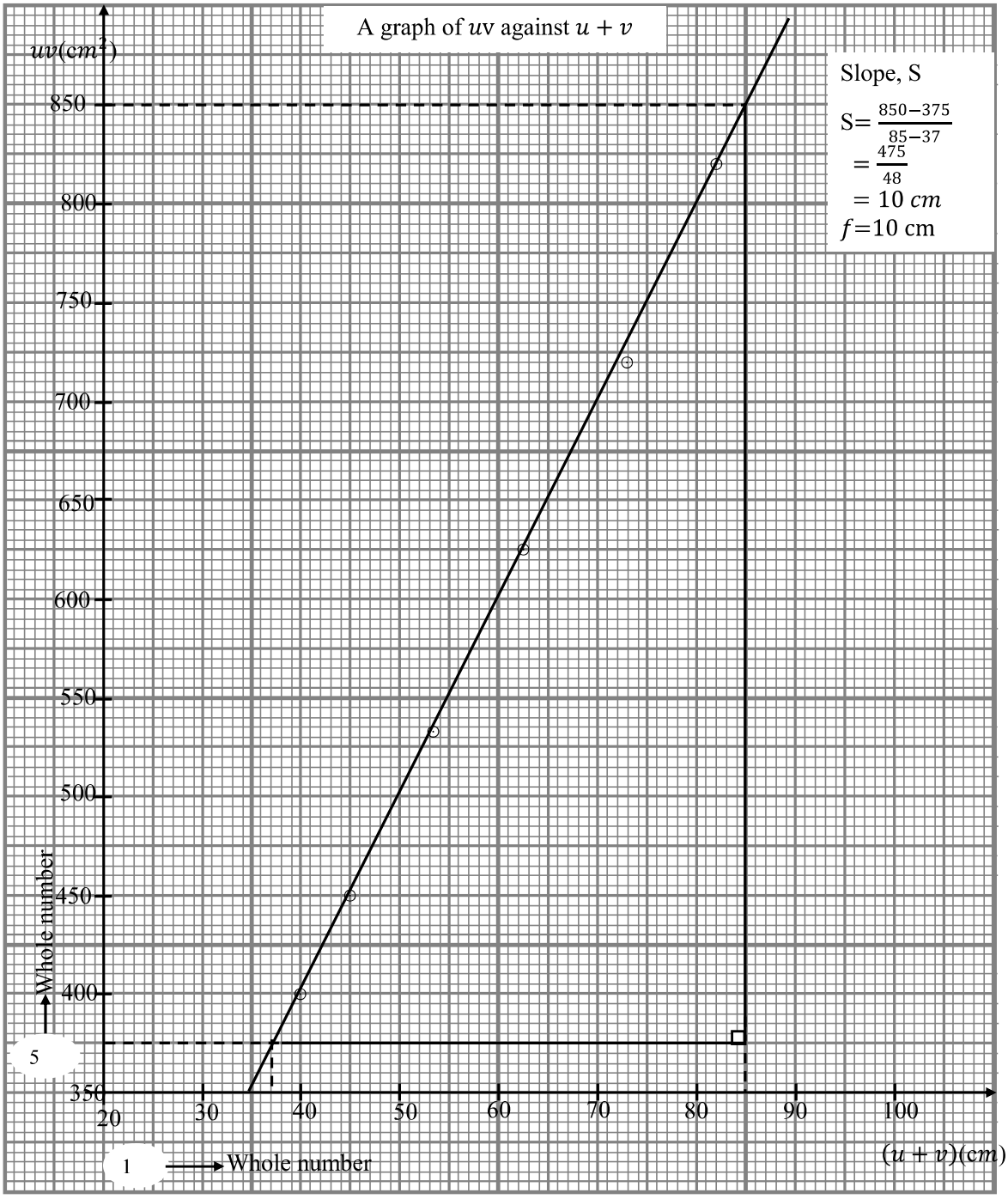
| **Source of error** | **Precaution** |
|---|---|
| Parallax error when reading the metre rule | Making the eye sight perpendicular to the reading being taken |
| Too much light in the room | Switching off the lights |
| Too little light produced by torch bulb | Increasing emf by connecting two cells in series |

| $u(cm)$ | v( cm) | $(u+v)(cm)$ | $uv(cm^2)$ |
|---|---|---|---|
| 20.0 | 20.0 | 40.0 | 400 |
| 30.0 | 15.0 | 45.0 | 450 |
| 40.0 | 13.3 | 53.3 | 532 |
| 50.0 | 12.5 | 62.5 | 625 |
| 60.0 | 12.0 | 62.0 | 720 |
| 70.0 | 11.7 | 81.7 | 819 |

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$\boldsymbol{f = 10cm = 0.10m}$ $\quad$ $\boldsymbol{P = \frac{1}{f(m)} = \frac{1}{0.10} = +10D}$

**Conclusion**

The power of the lens is $\boldsymbol{+10D}$ hence the hypothesis is accepted

**Recommendation**

I recommend that the lens be used for eye correction

**MARKING GUIDE PREPARED BY**

**MR NYARUHUMA THOMAS 0703240480**

**PROUD OF FORT PORTAL SS AND SKYVILLE HIGH SCHOOL**

535/1
**PHYSICS**
Paper 1
**Jul/Aug.2026**
2½ hours

**UGANDA TEACHERS' EDUCATION CONSULT (UTEC)**

**Uganda Certificate of Education**

PHYSICS

**Paper 1**

Theory

**2 hours 30 minutes**

**INSTRUCTIONS TO CANDIDATES:**

*This paper consists of two Sections A and B. It has seven examinations items*

*Section A has three compulsory items.*

*Section B has two parts: I and II. Answer one item from each part.*

*Answer five items in all*

*All responses must be written in the booklets provided*

## SECTION A (COMPULSORY)

**Item 1**

During the enforcement of a Uganda trade order, an L.C.1 defense personnel was instructed to inform traders that street vendors and buildings located at least 0.025 km away were to be demolished. The personnel used a hired megaphone and a motorcycle equipped with a side mirror that continuously located distant objects as he moved. While on the tarmac road, the personnel observed that the surface appeared to be a glittering pool of water which kept disappearing as he approached it, which made him wonder greatly. While communicating, he bypassed a large workshop building and began hearing his own spoken words repeated at intervals of 0.175 seconds. The personnel stopped the megaphone using a remote control, but upon switching it back on and communicating, the sound repeated again and only cleared when he moved away from the structure. Consequently, he felt the megaphone had mechanical issues. The owner of the building heard the announcement, although he could not clearly see distant objects, and was in fear of the demolition of his structure.

**Hint:**

- Speed of sound in air is 320 $ms^{-1}$
- Ultraviolet radiation is characterized by high frequency.

***Task:***

*Help:*

a) Clear the personnel's wonder regarding the glittering pool of water on the road.

b) Using sketch ray diagrams, explain to the owner of the workshop his particular eye defect and its correction.

c) Explain the side effects of prolonged exposure to the rays used by the remote-control device that the personnel used to switch off the megaphone.

d) Inform the owner of the structure regarding whether his building meets the demolition criteria.

e) Using a clear ray diagram, explain why the motorcycle mirrors located those distant objects and discuss the challenges of using them.

f) Give more applications of such mirrors in daily life.

**Item 2**

A nearby mine in a certain region of Uganda was suspected of discharging dangerous radioactive materials into the nearby swamp water that supports community activities. A study was carried out, and the results are presented below:

| Count rate (per hour) | 625 | 575 | 500 | 450 | 400 | 300 | 250 |
|---|---|---|---|---|---|---|---|
| Time (hours) | 0 | 10 | 30 | 46 | 60 | 90 | 118 |

The study also showed that the swamp should remain out of reach to the community until the count rate drops to 220 counts per hour; this finding was not received in good faith by the community. It was further suggested that the whole community should undergo individual medical check-ups if the half-life of the substance is above four (4) days.

The community also alleged that the mine consumes too much electricity, which affected their local supply. However, the miners defended themselves by stating that they have a plan to produce more reliable electricity through the nuclear fission of uranium, as represented by the following equation, though the community still doubted it:

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{144}_{x}Ba + ^{y}_{36}Kr + 3^{1}_{0}n + energy$$

**Task:**

a) Help the community members to:
   (i) Determine the estimated number of days the community members need to keep away.
   (ii) Know whether they will be subjected to further individual medical tests.

b) Kindly inform the community members to help them appreciate the resolution to stay away for the determined duration.

c) Using a well-labeled diagram, explain the possibility of producing electricity from the nuclear fission process described above.

d) Present a balanced nuclear equation by finding the values of x and y.

e) Explain the uses and international regulations regarding the use and control of such radioactive materials today.

**Item 3**

In February 2026, the USA and Israel engaged in a military conflict with Iran in the Middle East, during which Iran's Supreme Leader was assassinated at around 9:30 AM local time in Iran. The attack was closely monitored by the White House in the USA using space-based satellites, which asserted that the incident occurred at around midnight USA Standard Time. The Islamic Republic of Iran retaliated by launching sophisticated missiles at the State of Israel that penetrated the Israeli defense system, since the missiles traveled for almost 9 minutes at a hypersonic speed of Mach 13. As the war escalated, the entire planet Earth within the Solar System was affected, particularly as fuel prices skyrocketed.

**Support Material:**

- One Mach speed is equal to the speed of sound in air.
- Speed of sound in air is 330 $ms^{-1}$.

**Task:**

a) Explain the cause for the difference in the recorded time of occurrence of the attack that claimed the life of the Supreme Leader.

b) How was the White House able to observe and follow the attacks in such a distant country?

c) Estimate the distance between Iran and Israel based on the missile flight data.

d) Discuss other components of the Solar System.

e) Give several other applications of the satellite device setup that enabled the White House to follow up on the attack.

## SECTION B

### Part I

*Respond to **one item** from this part*

**Item 4**

John hired a special car to travel to school, which charged Ugsh 5,000 per kilometer on board. While loading luggage, John placed his bags on the top rack, but the driver changed the arrangement and put them in the lower cabin, which John did not like. As they started the journey, John hit his head hard against the seat when the car suddenly moved. He alarmed the driver, who stopped the car immediately, causing John to hit the forward seat. The driver then started slowly from rest and accelerated at a rate of 0.1 $m/s^2$ for 4 minutes. The driver continued at this steady speed for 10 minutes and bypassed an automatic penalty billing machine that fines every driver who exceeds a speed of 90 km/h for over speeding, and then decelerated uniformly to arrive at the

school in 3 minutes. The candidate offboarded and was charged Ugsh 155,520, which he doubted, claiming the driver had overcharged him beyond the agreed tariff. The candidate also developed a severe headache; when taken to the sickbay, the nurse attributed it to either fatigue or malaria. She checked his temperature, which read **310.6 K**. The nurse required that they pay for a further malaria test if the temperature exceeded **37.8 °C**.

**Task:**

Help;

a) (i) Explain the physical effect experienced by John when the car started and stopped, and how it could be limited.

(ii) Give applications of this effect in daily life.

b) Inform the driver whether he was billed for over speeding by the automated machine.

c) Help John appreciate the driver's decision regarding the position of the luggage.

d) Determine whether the candidate was required to pay for the further malaria test based on his temperature.

e) With reference to a sketch velocity-time graph, determine if the fare charged by the driver was ideal based on the distance covered.

✓ **Item 5**

Senior 1 learners set off for a study tour to the source of a certain river. While traveling in a saloon car to the site, a speeding Sinotruk bypassed them, and they experienced a force pulling them towards the truck, which made them fear greatly for their lives. On reaching the site, two girls, A and B, rushed to a seesaw where Girl A, of mass 46 kg, was seated at 2.5 m from the center pivot. Girl B, who weighed 600 N, wanted to balance the seesaw on the other end but did not know the exact distance where she should sit. They then witnessed some ice floating on water and also noticed that the river was flowing fastest at narrow sections compared to wider sections, which puzzled them. They continued to a dam where their teacher cautioned them that swimming in a dam is only safe if the depth does not exceed 4 m. The tour guide informed them that the dam had thicker walls at the bottom to counteract a hydrostatic pressure of around $6.5 \times 10^4$ N/m$^2$, and advised that they should not risk moving near the water because the dam appeared shallower than its actual depth.

**Hint:**

- Density of water is 1000 kgm$^{-3}$
- Acceleration due to gravity is 10 ms$^{-2}$

**Task:**

*Help*

a) Explain to the learners the cause of the force that caused their fear while on board next to the moving truck.

b) Determine the distance Girl B should sit from the pivot to balance the seesaw, and state the conditions necessary for it to remain in this equilibrium state.

c) Explain the anomalous behavior of ice to the girls and explain any of its applications in daily life.

d) Inform them whether they will be safely allowed to swim based on the maximum depth rule and the given pressure threshold.

e) Explain any other applications of the knowledge of fluid pressure acting normally on surfaces (such as the bottom of a dam) in daily life.

## Part II

*Respond to **one item** from this part*

**Item 6**

A poultry farmer in Mityana is required to connect four brooder infrared bulbs, each labeled 250W, 240V, to warm chicks, and a security bulb to be used on the compound. These are connected to the mains electricity supply that charges Ugsh 760 per unit, with a fixed service fee of Ug.sh 3,500 per week. He has two switches and a junction box to connect the bulbs, as well as a socket for a speaker used for advertising. The farmer needs to limit electricity consumption by minimizing the hours of use of the security bulb and the brooder bulbs as much as possible since they are highly energy-consuming. The farmer has been using a local heating charcoal stove at a cost of Ugsh 65,000 per bag of charcoal per week, and is not sure whether he will be saving money by switching to the brooder bulbs. He plans to use the brooder bulbs for 10 hours daily and strictly use the security light during the hours of darkness daily (assumed to be 12 hours).

**Task:**

As a learner of physics, help the farmer to:

a) make an informed decision between using the infrared bulbs and the charcoal stove by comparing costs.

b) make a safe and efficient circuit diagram layout for the above appliances in reference to costs, safety, and efficiency.

c) explain how the advertising speaker device works.

d) automatically switch on and off a compound bulb effectively in response to darkness.

e) give more applications of the rays used by brooder bulbs in daily life.

**Item 7**

In an effort to connect the Karamoja region to the national electricity grid from the River Nile, local residents were told to clear and vacate a stretch of land which was a minimum of 10 feet from the grid path. The transmission line is expected to carry a voltage of around 40 kV, which would later be stepped down to 14 kV using a transformer with 400 turns in its primary coil. The locals disagreed, claiming that all that land could not be left unutilized just because of a mere electric grid pathway. The officials then requested the Chairman L.C.1 of one of the villages to call for an awareness meeting. The chairman used a bicycle to ride around the village mobilizing residents. He tried turning on the headlight, but it did not work; he was told by a mechanic that the dynamo had internal technical issues, which was the source of light. The Chairman wondered greatly how the bicycle dynamo works to produce light without any dry cell battery.

**Task:**

a) Help convince the locals to appreciate the safety need to leave the demarcated grid space unutilized.

b) Kindly clear the chairman's wonder by explaining how a bicycle dynamo generates electricity.

c) With reference to calculations, advise on the exact specifications (number of turns in the secondary coil) of the transformer required to step down the voltage as specified.

d) Using a well-labeled diagram, explain to the locals the path of grid transmission from the River Nile power station to their region.

e) Help locals understand why electric power is transmitted along the grid at such a very high voltage.

f) Discuss more applications of the device used to step down or step-up voltage in everyday technology.

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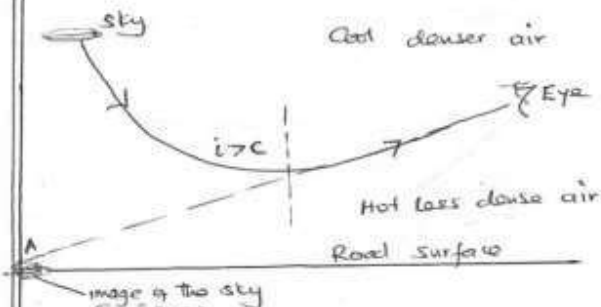
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0703240480 . JOM FORT PORTAL S.S OR SKYVILLE HIGH SCHOOL .

## ITEM ONE

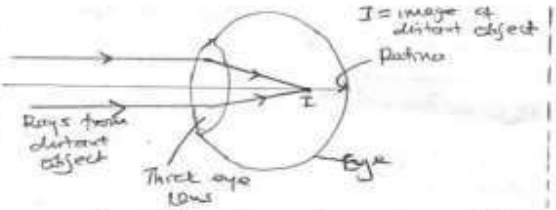
(a) The glittering pool of water on the road was a mirage . It is an image of the sky formed on the road by continuous refraction and total internal reflection.




i = angle of incidence
c = critical angle.

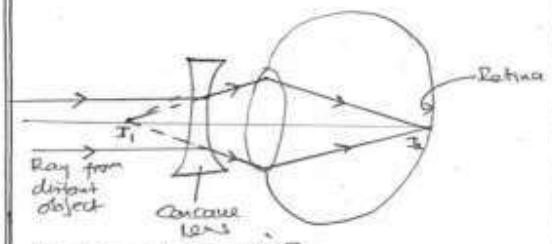
- During a hot day, air layers near the road are heated by the road and the sun .
- The air layer become hotter and less dense than air layers above .
- Light rays from the sky are continously refracted away from the normal and the angle of incidence increases .
- When the angle of incidence exceeds critical angle, total internal reflection occurs
- The rays fall into the eye of the observer and appear to be coming from the road at A .
- The image of the sky that appears like a pool of water is formed at A .

(b) The eye defect is called short Sightedness (myopia).




- The eye lens becomes too thick hence its focal length reduces .
- The lens forms an image of a distant object in front of the retina .
- This makes the image unclear / blurry

Correction of the eye defect




$I_1$ = Image first image
$I_2$ = Final image

- The defect is corrected by wearing spectacles containing Concave / diverging lenses
- Rays from a distant object are diverged by the concave lens hence appear to be coming from $I_1$ (far point) which is closer to the eye .
- The eye lens then focusses the rays onto the retina .
- The final image is clear .

(c) Effects of prolonged exposure to high frequency radiations . The best radiation for a megaphone remote are radiowaves . High frequency ionizing radiations such as Ultraviolet carry enough energy to break chemical bonds and knock out electrons from atoms directly damaging cellular structures and DNA by this leads to the following effects .

- Can cause cancer such as leukemia, skin cancer and thyroid cancer
- Can cause skin burn and hair loss
- Can cause eye cataract
- Can cause suppression of immune system
- Can cause sterility

(1)

(d) Least distance for demolition to occur = 0·025km
Echo interval time interval $t$ = 0·175 seconds
Speed of sound in air $V$ = 320 ms$^{-1}$
Let the distance of the structure be $d$

$V = \frac{2d}{t}$

$320 = \frac{2d}{0.175}$

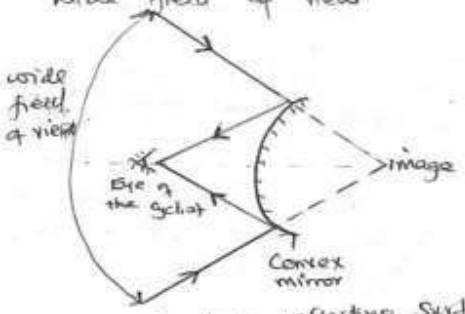
$d = 28m$

$= \frac{28}{1000} = 0.028km$

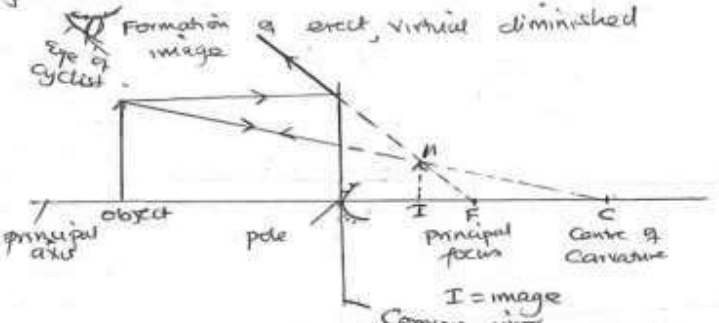
The structure is located 0·028km away which is greater than 0·025km. It should not be demolished.

(e) Motocycle mirrors are convex (diverging) mirrors. They are able to locate distant objects because they have a wide field of view and form erect images regardless of object distance.

Wide field of view




Formation of erect, virtual diminished image




Convex mirrors have reflecting surfaces that curve outwards. Rays from a broad area are incident on it

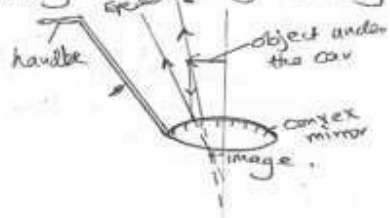
- A ray parallel to the principal axis is dw divergas off after reflection and appears to be coming from F.
- ~~May through Cen~~
- A ray appearing to pass through C is normal to the mirror surface and is reflected along the same path.
- The rays appear to meet at M forming a virtual, diminished image ~~at~~.

Challenges of using Convex mirrors

- Images are diminished which can lead to misjudgement.
- Reduced details and clarity

(f) More applizations of Convex mirrors

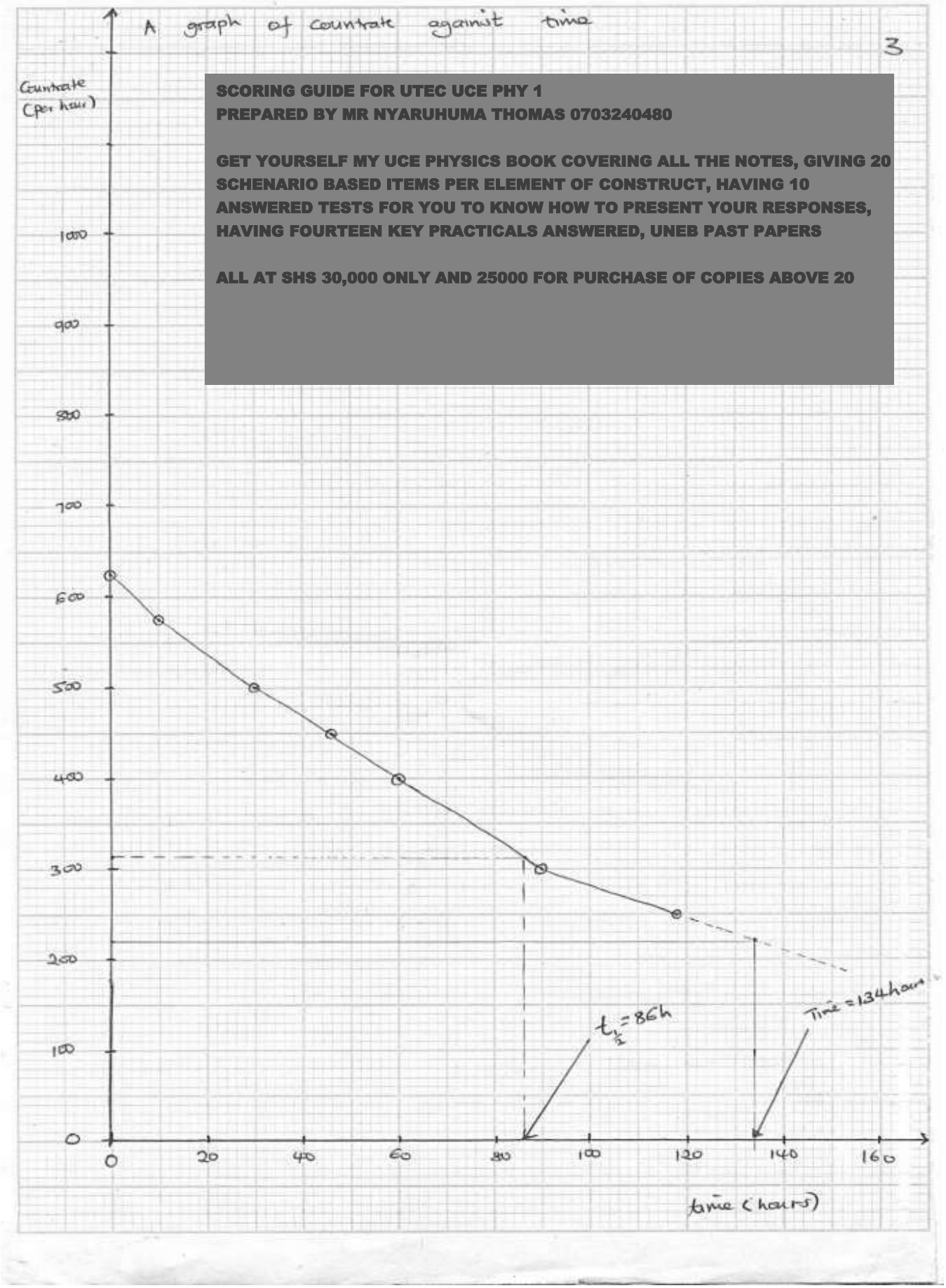
- They are used by Security to inspect under vehicles




- The observer is able to see the image of the object under the car by holding the handle so that the mirror lies beneath the car.

- They are used in monitoring Supermarkets. This is because of their wide field of view that enables each mirror to cover a large area of the Supermarket. They also give erect upright images for all objects in front for proper recognition

- They are used as dentist mirrors. When a tooth is placed at a distance less than focal length from the pole, the mirror gives a magnified imag




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ITEM TWO

(a) i From graph

Time = 134h $= \frac{134}{24} = 5.5833$ days.

The Community members need to keep away for 5.5833 days.

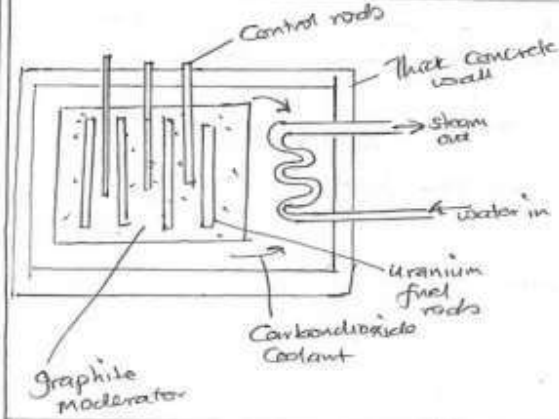
~~OR~~ 625

(ii) Half life, $t_{\frac{1}{2}} = 86h = \frac{86}{24} = 3.5833$ days.

The half life is below four days. They should not be subjected to further individual medical tests.

(b) Radioactive materials in the nearby Swamp ~~produce~~ decay Spontaneously and produce ~~radia~~ ionizing radiations that include alpha particles, beta particles and gamma rays. These radiations are dangerous to the human health in the following ways.

- they can cause genetical mutation
- they can cause leukemias
- they can cause sterility
- they can cause eye cataract
- they can reduce body immunity




- Uranium-235 is bombarded with neutron to produce U-236 which is extremely unstable.
- Uranium-236 undergoes nuclear fission to produce heat energy.
- Heat is absorbed by water which turns into steam.
- steam runs steam turbines and the steam turbines run the generator to produce electricity.
- the electricity is fed into the power grid for transmission.

(d) $^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{144}_{x}Ba + {}^{y}_{36}Kr + 3\,{}^{1}_{0}n$ + energy.

$235 + 1 = 144 + y + 3\times1 \Rightarrow y = 89$

$92 + 0 = x + 36 + 3\times0 \Rightarrow x = 56$

The balanced equation is

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n + \text{energy}$$

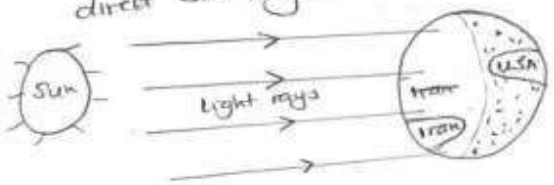
(e) uses and International regulations regarding use and control of radioactive materials.

- No one or group is allowed to own, store or use radioactive materials without licence for from a regulatory body. This helps to avoid risks of exposing the public to the radiations.
- Importation or exportation of radioactive materials should be through secure technical infrastructure to ensure global nuclear security.
- Workers in radioactive environments should observe safety measures to avoid health hazards.

* Unregulated disposal of radioactive wastes to into public water sources is prohibited to avoid health hazards in the public.

ITEM THREE.

(a) Cause of the difference in time recorded
- It was 9:30am (day) in Iran and midnight in USA because of rotation of the earth / spinning of the earth about its axis
- Iran was directly facing the sun and was receiving sunlight hence experienced day night time
- USA was facing away from the sun and was not receiving direct sun light hence experienced night time




(b) How the White house was able to observe and follow the attacks
- This was through use of Satellites.
- The space based infrared Sensors detect the sudden thermal flash produced by missiles and electro optical Cameras capture images and videos of troops.
- The raw data is processed into digital signals and transmitted to the ground station.
- The ground station uplinks the signals to the satellite which amplifies the signals and retransmits them downwards to the receiver.
- The signals are decoded and interpreted.

(c) V = Mach 13 = 13 x 330 = 4290 ms$^{-1}$
t = 9 minutes = 9 x 60 = 540 seconds
Distance, d = Vt = 4290 x 540 = 2316600 m
= $\frac{2316600}{1000}$ = 2316.6 km
The distance between Iran and Israel is 2316.6 km

(d) Other Components of the Solar System
- The Sun - It constitutes over 99% of the total mass of the Solar System
- is a luminous sphere made up of hydrogen and helium
- It is a main Sequence G-type star.
- Its immense gravitational pull holds other components of the Solar System
- The planets - It has eight planets namely mercury, Venus, earth, mars Jupiter, Saturn and Uranus devided into inner and outer planets
- The planets orbit the sun
- Dwarf planets - These are all celestial bodies that orbit the sun, are spherical but do not clear neighbourhood of debris.
They include pluto, Eris ~~and~~ Ceres and Eris

(c) other applications of Satellites.
- They are used for Communication
- They are used for weather forecasting
- They are used for navigation
- They are used for scientific research.

Other Components of the Solar System
* Moons
* Rings
* Asteroids

## SECTION B PART ONE

ITEM FOUR

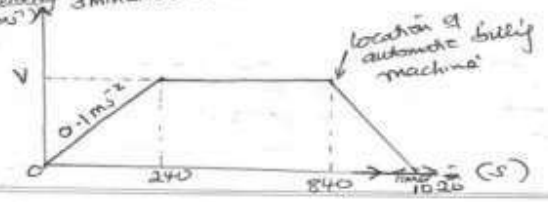
a(i) When the car suddenly started, John jerked backward due to inertia. Initially both John and the car were at rest. According to Newton's first law, when the car suddenly started, John remained at rest since the engine force that started the car did not act on John. John jerked backward and hit his head hard on the sit.

* When the car suddenly stopped, John jerked forward due to inertia. John and the car had attained the same speed. According to Newton's first law, when the car suddenly stopped, John continued in his state of uniform motion in a straight line due to inertia. This is because the braking force applied to the car did not act on John hence he jerked forward.

How this effect could be limited
* starting the car with low acceleration and this also stoping it with low deceleration. This reduces the inertia force on John
* Wearing an elastic seat belt. This provides and external force to overcome inertia force to prevent jerking off the seat. It is elastic to maximise the time of action of the force inorder to reduce impulsive force for less pain to be felt.

(ii) Applications of inertia
- Passengers wear seat belts to overcome inertia incase of a sudden stop.
- Athletes Long jumpers begin with running to develop enough inertia or momentum to jump high.
- Sprinters do not stop instantaneously after crossing the line finish line. The decelerate slowly through some distance to come to rest safely.
- When a dirty carpet is suspended and struck with a stick, fabric moves forward while dust particles tend to remain at rest hence seperating

(b) 4 minutes = 4×60 = 240 seconds
10 minutes = 10×60 = 600 seconds
3 minutes = 3×60 = 180 s




Finding velocity V
u = 0
v = v
a = 0·1
t = 240

v = u + at
= 0 + 0·1 × 240
= 24 ms⁻¹.

Speed = 24m / 1S = 24/1000 ÷ 1/3600
= 24/1000 × 3600/1
= 86.4 kmh⁻¹

The speed is below 90 kmh⁻¹. He was not billed for overspeeding.

(c) Why the driver placed the luggage in the lower cabin instead of top rack.

- When luggage is placed on top rack, the centre of gravity of the car rises. This reduces the stability of the car and increases chances of toppling.
- When luggage is placed in the lower cabin, centre of gravity of the car lowers thus increasing stability and reducing chances of toppling.

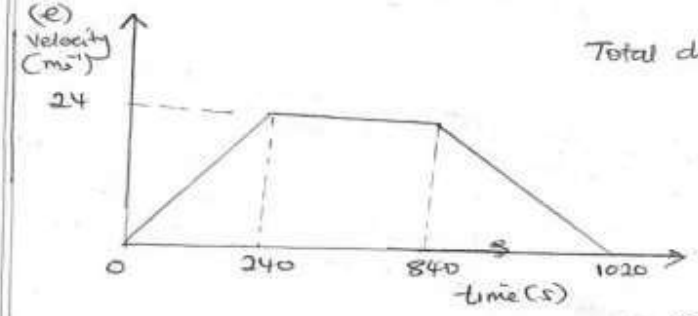
(d) $\theta(°C) = T(K) - 273$

$= 310.6 - 273$

$= 37.6\ °C$

The temperature is below 37.8°C. The candidate was not required to pay for further malaria tests.

(e)




Total distance = area under the graph

$= \frac{1}{2} \times 240 \times 24 + 600 \times 24 + \frac{1}{2} \times 180 \times 24$

$= 2880 + 14400 + 2160$

$= 19440\text{m}$

$= \frac{19440}{1000} = 19.44\text{ km}$

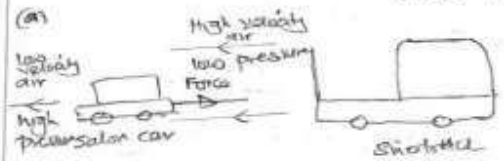
Charge = $19.44 \times 5000$ = shs 97200

Excess charge = 155520 − 97200 = shs 58320

The amount charged was not ideal. There was an excess charge of shs 58320

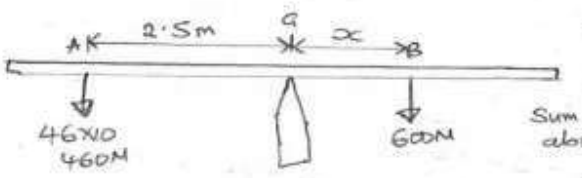
ITEM FIVE.

(a)




- The effect is called Suction effect. It is also explained using Bernoulli's principle.
- When the speeding shotmuck bypasses the Salon car, air moves faster between the track and the car than behind the Salon car. From Bernoulli's principle, pressure is higher behind the Salon car than between the Salon car and shotmuck. The pressure difference causes a force towards the low pressure region thus the car experiences a force pulling it towards the shotmuck.

(b)




Assumption
- The pivot is at the centre of gravity of the see saw

At equilibrium

Sum of clockwise moments about the pivot = Sum of anticlockwise moments about the pivot

$600x = 460 \times 2.5$

$x = 1.9167\text{m}$

Girl B should sit at 1.9167m from the pivot for the see saw to balance.

(b) Conditions necessary for the seasaw to remain in equilibrium
- The resultant force on the see saw should be zero.
- The sum of clockwise moments about any point should be equal to the sum of the anticlockwise moments about the same point.

(c) Anomalous behaviour of ice
- When water is cooled below 4°C, it expands instead of contracting.
- The strong hydrogen bonds between water molecules decrease during freezing hence increasing the intermolecular spacing.
- This causes expansion leading to a reduction in density. This makes ice to float on more dense water.

Application of the anomalous behaviour
- It provides thermal insulation for survival of aquatic animals. The ice floating ice traps water inside water from escaping to the environment to keeping the temperature range suitable for aquatic life.
- Help in global climate regulation. (The Albedo effect). The floating ice reflects large amounts of sun radiation back into space. This prevents oceans from absorbing the radiation hence stablising global temperature.

(d) $p = 6.5 \times 10^4 Nm^{-2}$
$\rho = 1000\ kgm^{-3}$
$g = 10ms^{-2}$
Depth $h = ?$

$p = \rho hg$
$6.5 \times 10^4 = 1000 \times 10 h$
$h = 6.5m$

The dam is deeper than 4m. It is not safe for swimming.

(e) other applications of pressure acting normally.
- Elevated water tanks ~~reservoirs~~. ~~Water rese~~ supply water tanks are given enough elevation to maximise height(h) of water to generate enough hydrostatic pressure to push water through the pipes.
- Submarines and deep sea reseach vessels are made of thick high strength spherical hulls to withstand the ~~high~~ crushing pressure at the bottom of the sea.
- Intravenous (IV) infusion drips. In hospitals drip bags are hang at a certain height to generate enough pressure to overcome patients blood pressure.
- Barometers and manometers use the liquid height to measure pressure.

ITEM SIX

(a) Power of the bulbs = $4 \times 250 = 1000W = \frac{1000}{1000} = 1kW$

Energy / units per week = $P(kW) \times t(h)$

$= 1 \times 10 \times 7 = 70kWh$

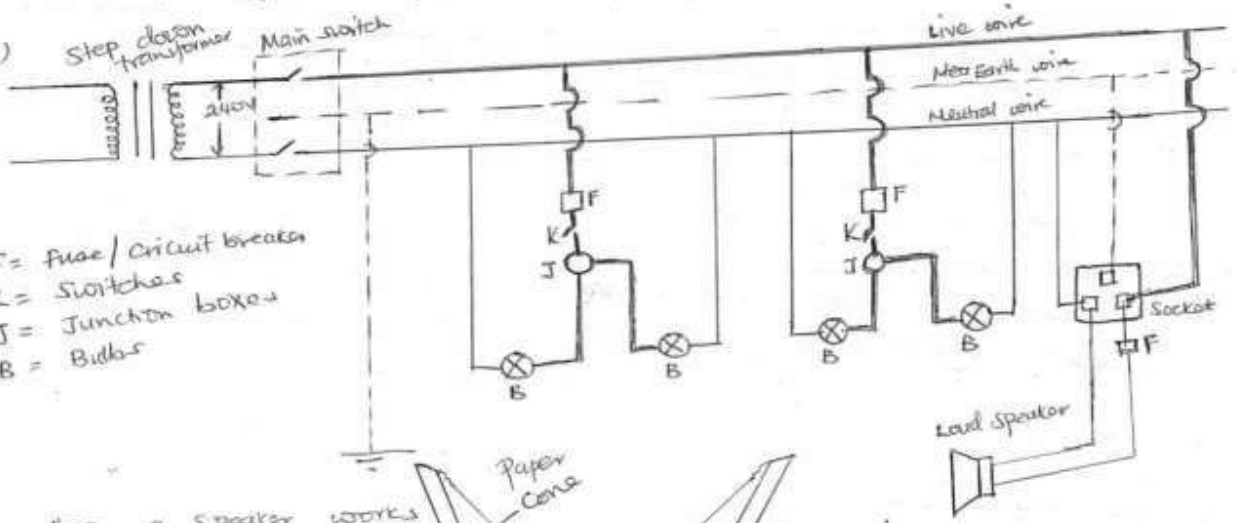
Cost per week $= 70 \times 760 + 3500 =$ ~~shs 65000~~ shs 56700

Cost of using charcoal per week = shs 65000

Saving $= 65000 - 56700 =$ shs 8300

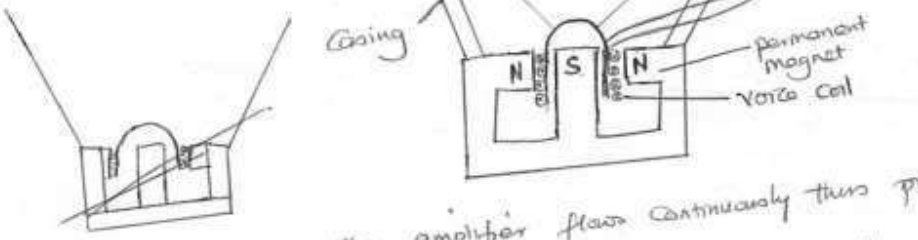
Use of infrared bulbs is cheaper than use of charcoal. He would save shs 8300 weekly by using infrared bulbs.

(b)



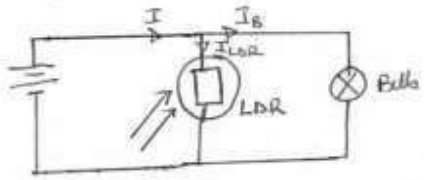

F = Fuse / Circuit breaker
K = Switches
J = Junction boxes
B = Bulbs

(c) How a speaker works




- Varying currents from the amplifier flow continuously thus producing varying force on the coil
- The coil vibrates with the same frequency and pushes the paper cone
- The paper cone vibrates thus setting large mass of air into vibration at the same frequency leading to production of sound.

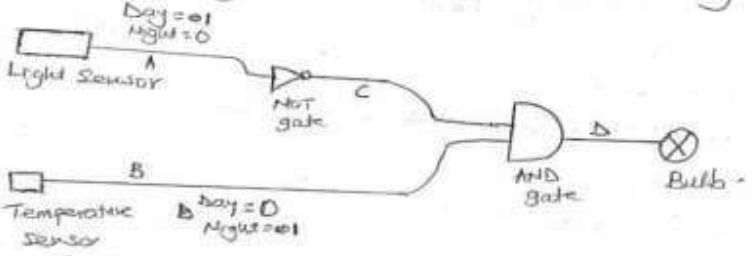
(d) Compound bulb can be automatically switched on or off using a light dependent resistor (LDR).




- During the day, light intensity is high. The resistance of the LDR is low. $I_{LDR}$ is high and $I_B$ is low keeping the bulb off.
- During the night, light intensity is low. The resistance of the LDR is high making current $I_{LDR}$ low and $I_B$ high. The bulb goes on.

OR

The security lights can be automated using logic gates with sensors.




Truth table

| A | B | C | D | Bulb |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | off |
| 0 | 1 | 1 | 1 | off on |
| 1 | 1 | 0 | 0 | off |
| 0 | 0 | 1 | 0 | off |

1 = Day
0 = Night

- During the day, the light sensor sends a ~~fal~~ true to the ~~Not~~ NOT gate which outputs a false. The temperature sensor also sends a false. This makes the AND gate to also send a false. The bulb keeps off.
- During the night, the light sensor sends a false to the NOT gate which outputs a true to the AND gate. The temperature sensor outputs a true. The AND gate outputs a true making the bulb to go on.

(e) Applications of infrared rays.

* IR remote control. IR LED produces infrared signals to encode binary data with specific commands such as changing channel. A photodiode or IR receiver detects the signals. The microprocessor decodes the signals and executes the command.
* Night vision equipment.
* Thermal imaging
* Security and intrusion detection

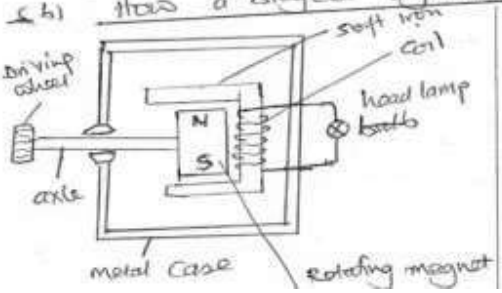
ITEM SEVEN.

(a) ~~High voltage~~ lines produce low frequency electromagnetic fields, also cause ~~ionization of~~ surrounding

(a) ~~why~~ Locals should not stay near high voltage power lines.

- ~~the~~ High voltage power lines produce low frequency electromagnetic fields that can cause childhood leukemia.
- High voltage power lines cause ionization of surrounding air hence producing noise due to corona discharge hence leading to discomfort
- High voltage power lines produce low frequency ~~pow~~ electromagnetic fields which cause interference of TV, radio and phone signals.
- In event of extreme weather such as high winds, the power lines may fall leading to fire hazards and electrocution.

(b) How a bicycle dynamo works




- It works on the principle of electromagnetic induction to convert mechanical energy into electrical energy.
- As the bicycle wheel rotates, it makes the driving wheel to rotate the magnet.
- This produces a changing magnetic field which includes an emf across the coil.
- An ac flows through the head lamp making it to produce light.

b

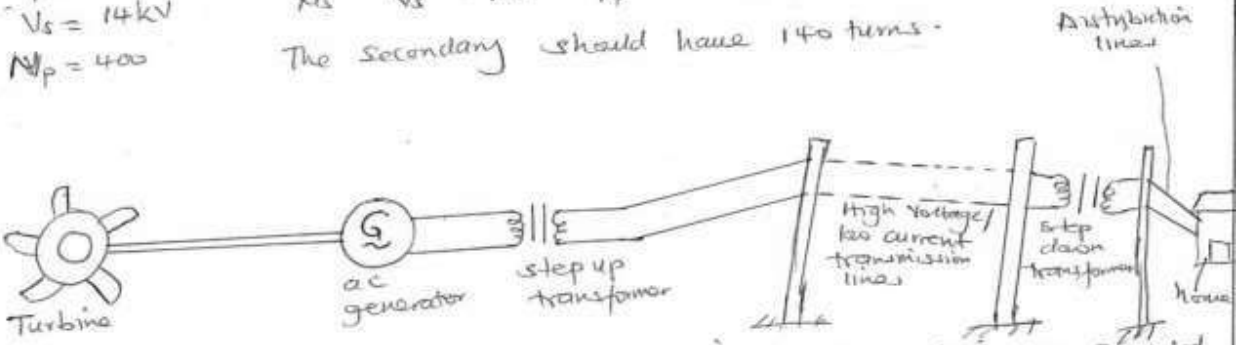
(c) $V_p = 40kV$ $\frac{N_p}{N_s} = \frac{V_p}{V_s}$ $\frac{400}{N_s} = \frac{40}{14}$ $N_s = 140$ turns

$V_s = 14kV$

$N_p = 400$

The secondary should have 140 turns.

(d)




- Water from the dam ~~runs the~~ rotates the turbines. The turbines are connected to the rotor of the ac generator making the generator coil to rotate. The generator produces low voltage alternating current.
- The ac voltage is stepped up using a step up transformer to minimise energy losses in form of heat produced by $I^2R$ mechanism during transmission
- ~~Power is then~~ The voltage is then stepped down to make it suitable for homes.

(e) Electric power is transmitted at high voltage so as to reduce the current in order to reduce ~~the~~ power losses in form of heat produced by $I^2R$ mechanism during transmission.

(f) Applications of transformers.

- used to step up voltage for long distance power transmission
- used in small electronics and chargers ~~to~~ e.g phone charger to step down voltage to very low suitable voltage


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